Activity

Group members:

Grad applications data

Using the grad applications data, you fit a model with GPA as the explanatory variables. Here is the regression output:

```
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -4.3576     1.0353  -4.209 2.57e-05 ***
gpa           1.0511     0.2989   3.517 0.000437 ***
```

And here is the variance-covariance matrix for the fitted model:

```
            (Intercept)        gpa
(Intercept)   1.0718814 -0.3076994
gpa          -0.3076994  0.0893229
```

Questions

1. Calculate the estimated log odds of admission for a student with GPA 3.8.

Fact: For two variables U, V and constants a, b:

$$Var(aU + bV) = a^2Var(U) + b^2Var(V) + 2abCov(U, V)$$

For a student with GPA 3.8, the estimated log odds is

$$\widehat{\log \text{odds}} = \hat{\beta}_0 + \hat{\beta}_1(3.8)$$

2. Using the fact above, and the variance-covariance matrix from the R output, calculate $Var(\widehat{\log \text{odds}})$ when GPA = 3.8.

3. From your answer in question 2, calculate $SE(\widehat{\log \text{odds}})$.

A 95% CI for the *true* log odds of admission for students with GPA 3.8 is

$$\widehat{\log \text{odds}} \pm 1.96 SE(\widehat{\log \text{odds}})$$

4. Using your answers to questions 1 and 3, calculate a 95% CI for the true log odds of admission for students with GPA 3.8.

5. Finally, transform your confidence interval from question 4 into a 95% CI for the true *probability* of admission for students with GPA 3.8.